

Moments

Draw every force on the rod. Take moments about the point you do not care about. Then resolve.
On the point of tilting, one reaction is zero.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Moments

01 - The moment of a force

force $\times$ perpendicular distance resultant moment

02 - A rod in equilibrium

two supports uniform rod $T_C = 2T_D$

03 - On the point of tilting

other reaction = 0 range of M

04 - Non-uniform rods

find the centre of mass two tilting cases

THE TWO EQUATIONS OF EVERY MOMENTS QUESTION

$\uparrow$ forces = $\downarrow$ forces clockwise moments =
anticlockwise moments

resolve vertically

about any point you choose

01 The Moment of a Force

Moment = force $\times$ **perpendicular** distance from the point to the line of the force. Units: N m. Always say clockwise or anticlockwise.

WHEN YOU SEE $\rightarrow$ YOU DO

“moment of the force about P ”

$\rightarrow F \times d_{\perp}$, with a direction

the distance is not perpendicular (angle θ)

$\rightarrow F \times d \sin \theta$ (or use the perpendicular component of F)

“resultant moment” / several forces

$\rightarrow$ choose a positive sense, add with signs

force passes through the point

$\rightarrow$ its moment is zero



moment about P
 $= F \times d \sin \theta$ (anticlockwise)

WORKED MODEL • RESULTANT MOMENT

Light rod: 4 N up at 1.2 m left of P ; 2 N and 1 N up at 1.2 m and 4.8 m right of P .

1 anticlockwise $+$: $-4(1.2) + 2(1.2) + 1(4.8)$

signs from the sense

2 $-4.8 + 2.4 + 4.8$

each $F \times d$

ANSWER 2.4 N m anticlockwise

IDEA 01

Practise: The Moment of a Force

Questions written to practise exactly this idea.

Q1 DR ESLAM Perpendicular distance

M1 · TOPIC 10 2 marks

A force of 12 N acts at the end of a 3 m rod, at 30° to the rod. Find its moment about the other end.

Q2 DR ESLAM Resultant with signs

M1 · TOPIC 10 3 marks

A light rod AB of length 4 m. Forces: 5 N down at A , 3 N down at B , 2 N up at the midpoint M . Find the resultant moment about M .

02 A Rod in Equilibrium

Two unknown forces, two equations. Take moments about one unknown so it disappears, then resolve for the other.

WHEN YOU SEE → YOU DO

“rests on two supports” / “held by two ropes”

→ two unknown reactions / tensions

“uniform rod”

→ weight at the **midpoint**

“the tension at C is twice the tension at D”

→ write both as T and $2T$ first

“in terms of g ” / masses in kg

→ forces are mg ; keep g or use 9.8

“state how you used ... rod / particle”

→ rod: stays straight · particle: weight at one point

WORKED MODEL · JUNE 2026

Rod 6 m, 20 kg, centre of mass 4 m from A; strings at P (0.5 m) and Q (4.5 m).

1 $M(P): T_Q(4) = 20g(3.5)$

moments about P

2 $T_Q = 17.5g$

solve

ANSWER $T_Q = 171.5 \text{ N}$

IDEA 02

Practise: A Rod in Equilibrium

My questions first, then real past papers.

Q3 DR ESLAM Moments about a support

M1 · TOPIC 10 3 marks

A uniform rod AB , 4 m, weight 60 N, rests on supports at A and C where $AC = 3$ m. Find both reactions.

Take moments about A first.

Q4 DR ESLAM Ratio of tensions

M1 · TOPIC 10 4 marks

A uniform rod AB , 6 m, mass 10 kg, hangs from ropes at A and at C . The tension at C is 4 times the tension at A . Find AC .

$AC =$ _____

PAST PAPERS

Q5 PAST PAPER Pearson IAL WME01 • June 2023 • Q4

M1 • TOPIC 10 12 marks

4.

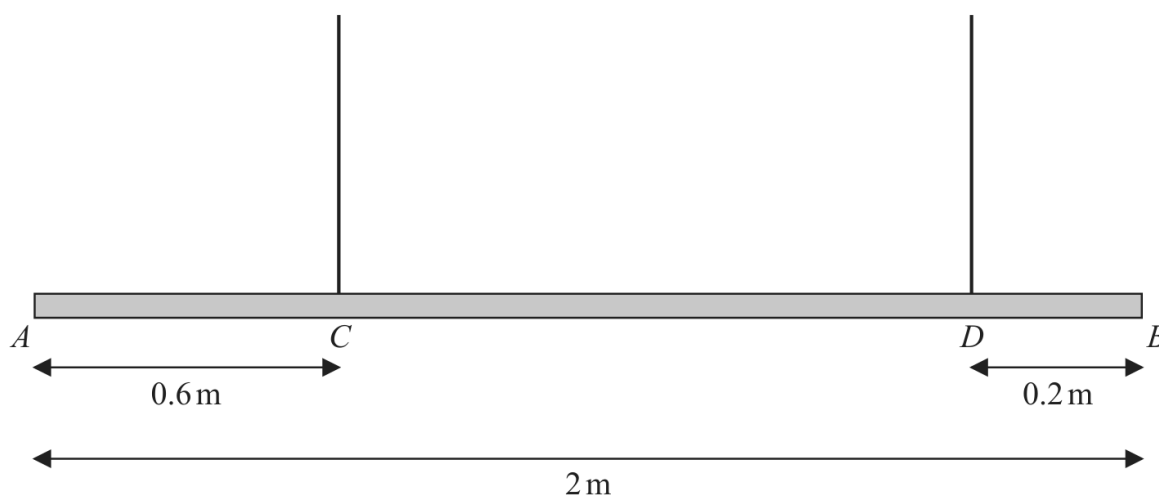


Figure 1

Figure 1 shows a beam AB , of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

A particle of mass pm kg is attached to the beam at A and the beam remains in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

- (a) Given that the tension in the rope attached at C is four times the tension in the rope attached at D , use the model to find the exact value of p .

(7)

The particle of mass pm kg at A is removed and replaced by a particle of mass qm kg at A .

The beam remains in equilibrium in a horizontal position but is now on the point of tilting.

- (b) Using the model, find the exact value of q

(4)

- (c) State how you have used the modelling assumption that the beam is uniform.

(1)

Q6 PAST PAPER Pearson IAL WME01 · October 2023 · Q1

M1 · TOPIC 10 5 marks

1.

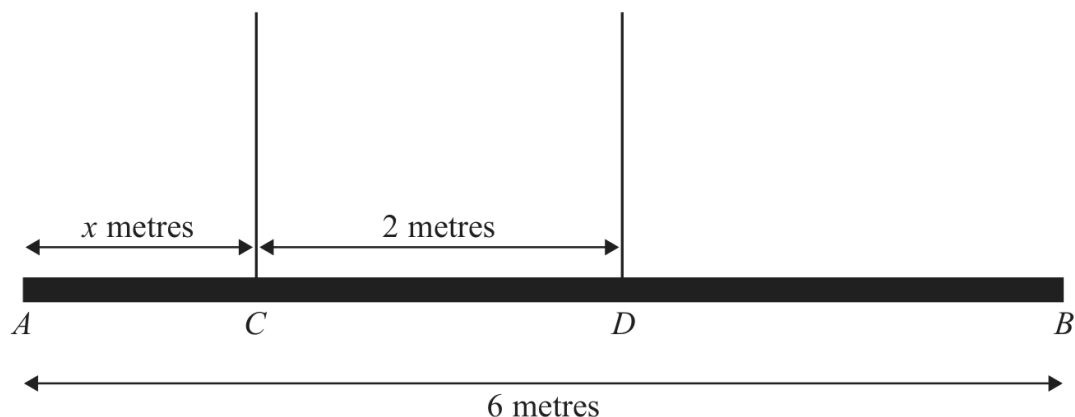
**Figure 1**

Figure 1 shows a beam AB with weight 24 N and length 6 m .

The beam is suspended by two light vertical ropes. The ropes are attached to the points C and D on the beam where $AC = x$ metres and $CD = 2\text{ m}$.

The tension in the rope attached to the beam at C is double the tension in the rope attached to the beam at D .

The beam is modelled as a uniform rod, resting horizontally in equilibrium.

Find

- (i) the tension in the rope attached to the beam at D .
- (ii) the value of x .

(5)

Q7 PAST PAPER Pearson IAL WME01 · January 2024 · Q5

M1 · TOPIC 10 10 marks

5.

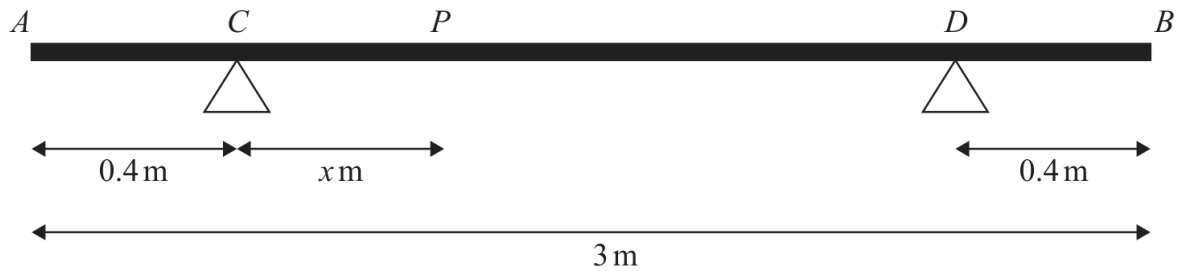


Figure 4

A beam AB has mass 30 kg and length 3 m.

The beam rests on supports at C and D where $AC = 0.4$ m and $DB = 0.4$ m, as shown in Figure 4.

A person of mass 55 kg stands on the beam between C and D .

The person is modelled as a particle at the point P , where $CP = x$ metres and $0 < x < 2.2$

The beam is modelled as a uniform rod resting in equilibrium in a horizontal position.

Using the model,

(a) show that the magnitude of the reaction at C is $(686 - 245x)$ N. (3)

The magnitude of the reaction at C is **four** times the magnitude of the reaction at D .

Using the model,

(b) find the value of x (4)

The person steps off the beam and places a package of mass M kg at A .

The package is modelled as a particle at the point A .

The beam is now on the point of tilting about C .

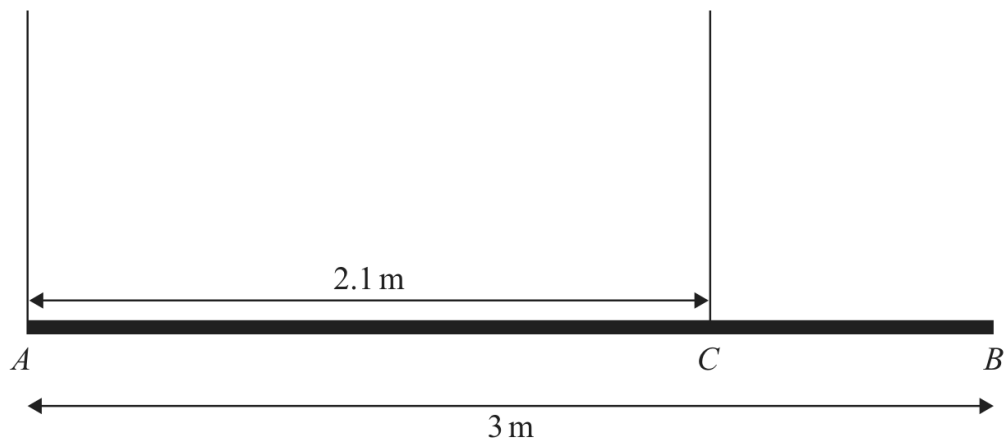
Using the model,

(c) find the value of M (3)

Q8 PAST PAPER Pearson IAL WME01 • January 2025 • Q4

M1 • TOPIC 10 10 marks

4.

**Figure 2**

A uniform rod AB has length 3 m and weight W newtons.

The rod is suspended by two light vertical ropes.

One rope is attached to the rod at A and the other rope is attached to the rod at C , where $AC = 2.1$ m.

The rod is in equilibrium in a horizontal position, as shown in Figure 2.

The tension in the rope at C is 350 N.

(a) Show that $W = 490$

(3)

A particle P of weight 210 N is attached to the rod at a distance d metres from A .

The tension in the rope at C is now 600 N.

The rod remains in equilibrium in a horizontal position.

(b) Find the value of d .

(3)

Particle P is removed from the rod.

A particle Q of weight X newtons is now attached at B .

The rod remains in equilibrium in a horizontal position and is now on the point of tilting.

(c) Find the value of X .

(4)

Q9 PAST PAPER Pearson IAL WME01 · October 2025 · Q1

M1 · TOPIC 10 7 marks

1.

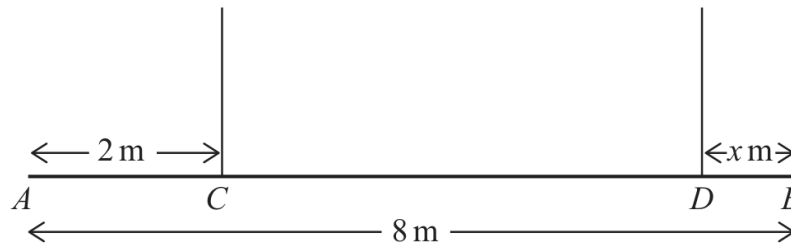
**Figure 1**

Figure 1 shows a sketch of a beam AB , with weight 240 N and length 8 m .

The beam is held in equilibrium in a horizontal position by two vertical ropes. The ropes are attached to the beam at the points C and D , where $AC = 2\text{ m}$ and $DB = x\text{ metres}$.

The beam is modelled as a uniform rod and the ropes are modelled as light inextensible strings.

The tension in the rope at D is 90 N .

(a) Show that $x = \frac{2}{3}$ (3)

The rope at C will break if its tension exceeds 183 N . The rope at D cannot break. A package of weight W newtons is now attached to the beam at A . The beam remains horizontal and in equilibrium.

The package is modelled as a particle.

It is given that the rope at C does not break.

(b) Find the greatest possible value of W . (4)

Q10 PAST PAPER Pearson IAL WME01 · January 2026 · Q4

M1 · TOPIC 10 9 marks

4.

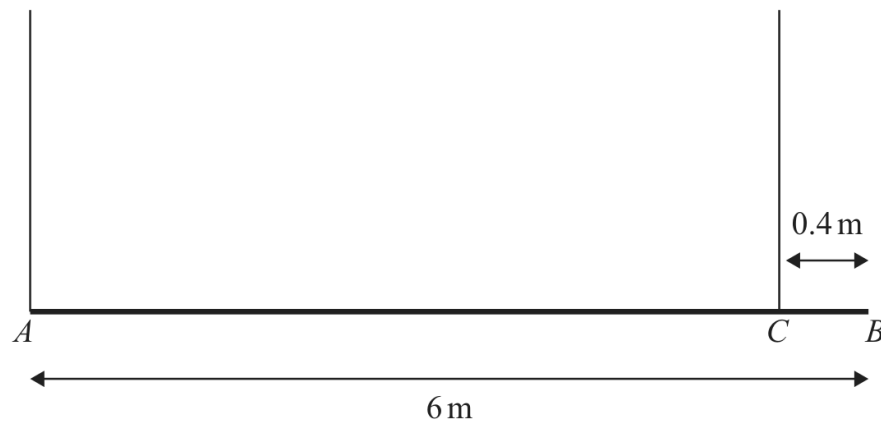
**Figure 2**

Figure 2 shows a uniform rod AB of length 6 m suspended by two light vertical ropes.

The first rope is attached to the rod at A .

The second rope is attached to the rod at the point C , where $CB = 0.4$ m.

The rod has mass 30 kg.

A particle of mass 20 kg is now attached to the rod at B .

The rod is in equilibrium in a horizontal position.

(a) Find

- (i) the tension in the rope attached to the rod at C ,
- (ii) the tension in the rope attached to the rod at A .

(6)

The particle of mass 20 kg at B is removed and replaced by a particle of mass M kg.

The rod remains in equilibrium in a horizontal position.

(b) Find the exact maximum value of M .

(3)

03 On the Point of Tilting

The rod is about to turn about one support. The other support has lifted: its reaction is **zero**.

WHEN YOU SEE → YOU DO

“on the point of tilting about D ”

→ reaction at the other support = 0 • moments about D

“largest / smallest mass so the rod stays horizontal”

→ it is on the point of tilting

“a rope is about to go slack”

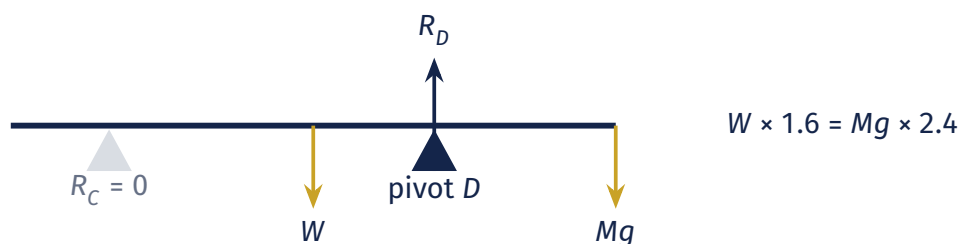
→ that tension = 0

“hence find the range of M ”

→ each reaction ≥ 0

overhangs a step / ledge

→ the edge is the pivot



WORKED MODEL • JANUARY 2022

Uniform beam 5 m on a step, edge D with $DB = 1.3$ m; a 30 kg boy at C ($CB = 0.5$ m) — on the point of tilting.

- 1 pivot D ; weight of the beam 1.2 m on the other side

tilting: A lifts

- 2 $M(1.2) = 30(0.8)$

moments about D

ANSWER $M = 20$ kg

IDEA 03

Practise: On the Point of Tilting

My questions first, then real past papers.

Q11 DR ESLAM Reaction becomes zero

M1 · TOPIC 10 3 marks

A uniform plank AB , 5 m, mass 20 kg, rests on supports at C ($AC = 1$ m) and B . How far from C , towards A , can a 40 kg child stand before the plank tilts?

Q12 DR ESLAM Range of a mass

M1 · TOPIC 10 4 marks

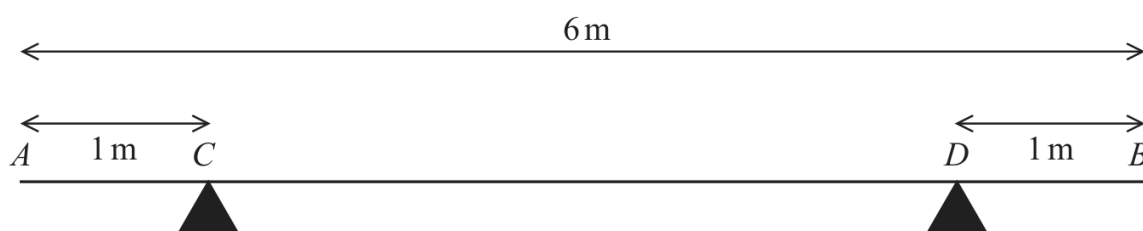
Uniform rod AB , 4 m, 10 kg, on supports 1 m from each end. A mass M kg hangs at B . Find the largest M so that the rod stays horizontal.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 • January 2021 • Q4

M1 • TOPIC 10 6 marks

4.

**Figure 2**

A metal girder AB has weight W newtons and length 6 m. The girder rests in a horizontal position on two supports C and D where $AC = DB = 1$ m, as shown in Figure 2.

When a force of magnitude 900 N is applied vertically upwards to the girder at A , the girder is about to tilt about D .

When a force of magnitude 1500 N is applied vertically upwards to the girder at B , the girder is about to tilt about C .

The girder is modelled as a non-uniform rod whose centre of mass is a distance x metres from A .

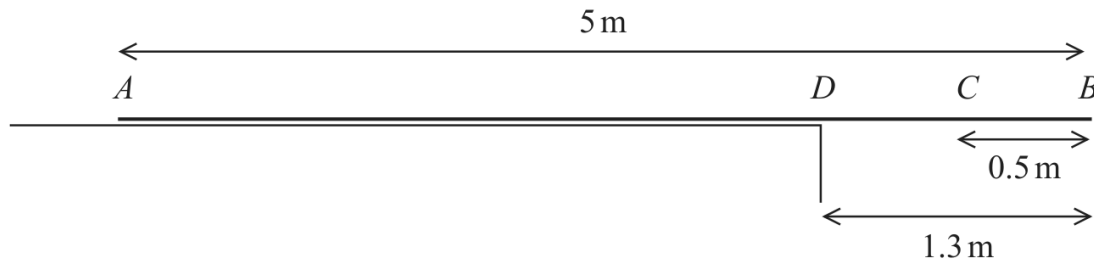
Find the value of x .

(6)

Q14 PAST PAPER Pearson IAL WME01 · January 2022 · Q3

M1 · TOPIC 10 7 marks

3.

**Figure 2**

A beam $ADCB$ has length 5 m . The beam lies on a horizontal step with the end A on the step and the end B projecting over the edge of the step. The edge of the step is at the point D where $DB = 1.3\text{ m}$, as shown in Figure 2.

When a small boy of mass 30 kg stands on the beam at C , where $CB = 0.5\text{ m}$, the beam is on the point of tilting.

The boy is modelled as a particle and the beam is modelled as a uniform rod.

(a) Find the mass of the beam.

(3)

A block of mass $X\text{ kg}$ is now placed on the beam at A .

The block is modelled as a particle.

(b) Find the smallest value of X that will enable the boy to stand on the beam at B without the beam tilting.

(3)

(c) State how you have used the modelling assumption that the block is a particle in your calculations.

(1)

Q15 PAST PAPER Pearson IAL WME01 · June 2022 · Q5

M1 · TOPIC 10 9 marks

5.

**Figure 2**

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D , where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

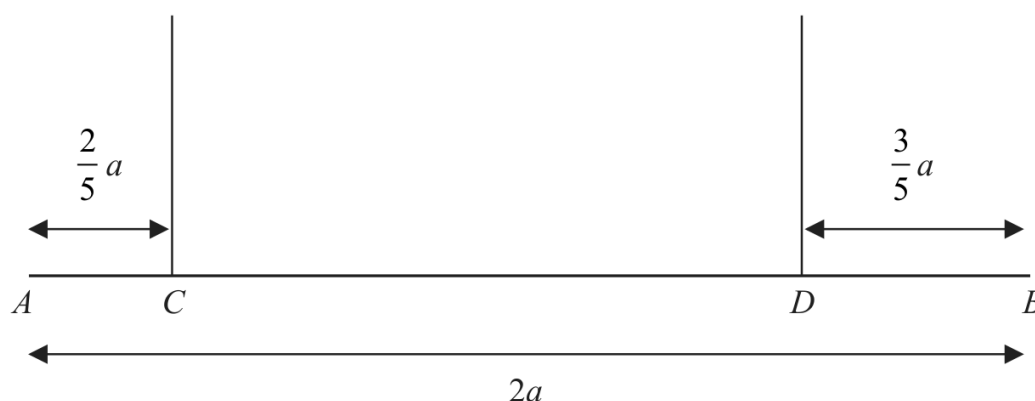
A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B . The rod remains horizontal and in equilibrium.

- (a) Find, in terms of M , the magnitude of the reaction on the rod at C . (3)
- (b) Find, in terms of M , the magnitude of the reaction on the rod at D . (3)
- (c) Hence, or otherwise, find the range of possible values of M . (3)

Q16 PAST PAPER Pearson IAL WME01 · October 2022 · Q2

M1 · TOPIC 10 6 marks

2.

**Figure 1**

A uniform rod AB has length $2a$ and mass M . The rod is held in equilibrium in a horizontal position by two vertical light strings which are attached to the rod at C and D ,

where $AC = \frac{2}{5}a$ and $DB = \frac{3}{5}a$, as shown in Figure 1.

A particle P is placed on the rod at B .

The rod remains horizontal and in equilibrium.

(a) Find, in terms of M , the largest possible mass of the particle P (3)

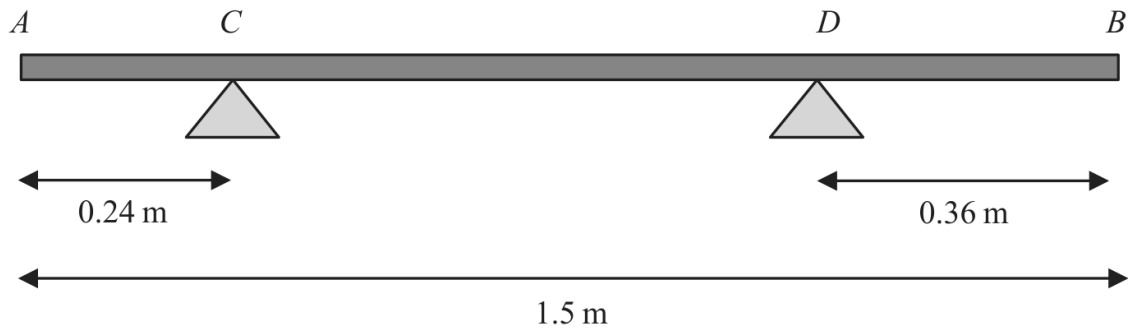
Given that the mass of P is $\frac{1}{2}M$

(b) find, in terms of M and g , the tension in the string that is attached to the rod at C . (3)

Q17 PAST PAPER Pearson IAL WME01 · January 2023 · Q4

M1 · TOPIC 10 8 marks

4.

**Figure 1**

A branch AB , of length 1.5 m, rests horizontally in equilibrium on two supports.

The two supports are at the points C and D , where $AC = 0.24$ m and $DB = 0.36$ m, as shown in Figure 1.

When a force of 150 N is applied vertically upwards at B , the branch is on the point of tilting about C .

When a force of 225 N is applied vertically downwards at B , the branch is on the point of tilting about D .

The branch is modelled as a non-uniform rod AB of weight W newtons.

The distance from the point C to the centre of mass of the rod is x metres.

Use the model to find

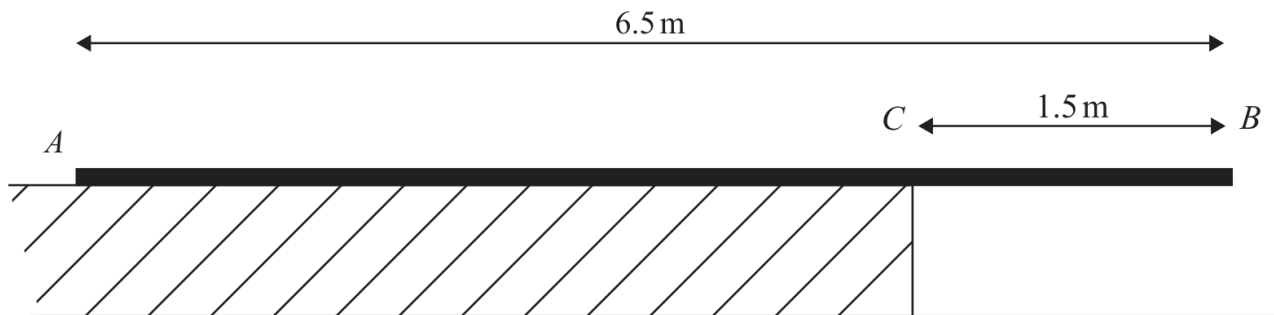
- (i) the value of W
- (ii) the value of x

(8)

Q18 PAST PAPER Pearson IAL WME01 · June 2024 · Q4

M1 · TOPIC 10 6 marks

4.

**Figure 2**

A non-uniform rod AB has length 6.5 m and mass 1.2 kg. The centre of mass of the rod is 3 m from A . The rod rests on a horizontal step and overhangs the end of the step C by 1.5 m, as shown in Figure 2.

The rod is perpendicular to the edge of the step.

A particle of mass 4 kg is placed on the rod at B and another particle, whose mass is M kg, is placed on the rod at D , where $AD = 0.5$ m.

The rod remains in equilibrium in a horizontal position.

(a) Find the smallest possible value of M .

(3)

The particle at B **and** the particle at D are now **removed**.

A new particle is placed on the rod at the point E , where $EB = 0.9$ m.

The rod remains in equilibrium in a horizontal position but is on the point of tilting about C .

(b) Find the magnitude of the force acting on the rod at C .

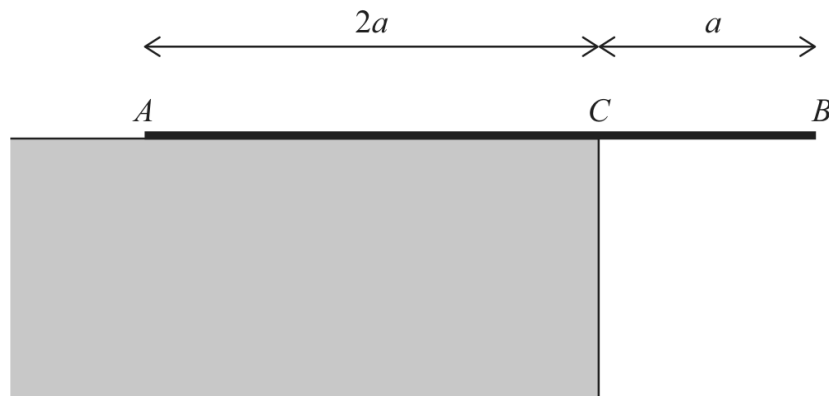
(3)

Q19**PAST PAPER**

Pearson IAL WME01 · June 2025 · Q5

M1 · TOPIC 10**6 marks**

5.

**Figure 3**

A non-uniform rod AB of length $3a$ rests in equilibrium on a horizontal ledge and overhangs the edge of the ledge at C .

The point C is such that $AC = 2a$ and $CB = a$, as shown in Figure 3.

The rod has weight W .

The distance of the centre of mass of the rod from A is d .

The rod is perpendicular to the edge of the ledge.

When a force of magnitude P , **acting vertically upwards**, is applied to the rod at B , the rod is on the point of tilting about A .

When the force applied at B is replaced by a force of magnitude $1.25P$, **acting vertically downwards** at B , the rod is on the point of tilting about C .

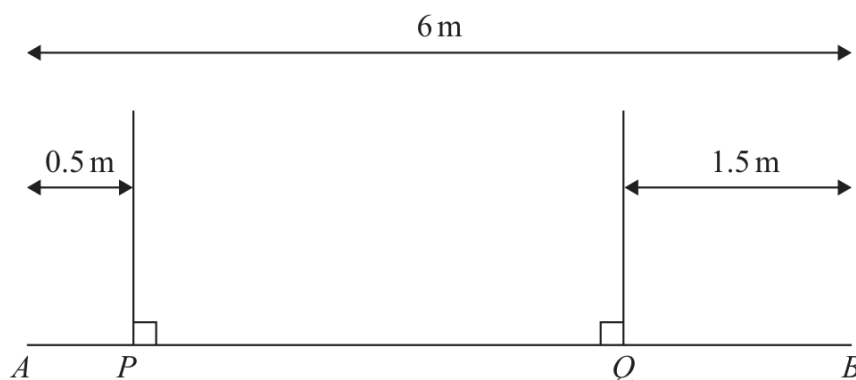
Find d in terms of a .

(6)

Q20 PAST PAPER Pearson IAL WME01 · June 2026 · Q2

M1 · TOPIC 10 6 marks

2.

**Figure 1**

A non-uniform rod AB has mass 20 kg and length 6 m. The rod is held in equilibrium in a horizontal position by two light vertical strings. One string is attached to the rod at P , where $AP = 0.5$ m and the other is attached to the rod at Q , where $QB = 1.5$ m, as shown in Figure 1.

The centre of mass of the rod is 4 m from A .

- (a) Find the tension in the string attached to the rod at Q . (3)

A particle is now attached to the rod at B .

The rod remains in equilibrium in a horizontal position but is on the point of tilting.

The tension in the string attached to the rod at Q is now X newtons.

- (b) Find the value of X . (3)

(Total for Question 2 is 6 marks)

04 Non-Uniform Rods: Find the Centre of Mass

The centre of mass is an unknown distance x . Two situations give two equations — solve them together.

WHEN YOU SEE → YOU DO

“non-uniform rod”

→ centre of mass at x from A — an unknown

“the reaction at D is three times the reaction at C ”

→ R and $3R$, then moments

two “point of tilting” situations

→ two equations → divide to cancel W

“show that $x = 1.38$ ”

→ finish with the exact number given

WORKED MODEL • JANUARY 2021

Girder 6 m, supports at 1 m and 5 m. 900 N up at A → about to tilt about D ; 1500 N up at B → about to tilt about C .

$$1 \quad 900(5) = W(5 - x)$$

about D , $R_D = 0$

$$2 \quad 1500(5) = W(x - 1)$$

about C , $R_C = 0$

$$3 \quad \frac{4500}{7500} = \frac{5 - x}{x - 1}$$

divide: W cancels

ANSWER $x = 3.5 \text{ m}$

IDEA 04

Practise: Non-Uniform Rods

My questions first, then real past papers.

Q21 DR ESLAM Find x from a ratio

M1 · TOPIC 10 4 marks

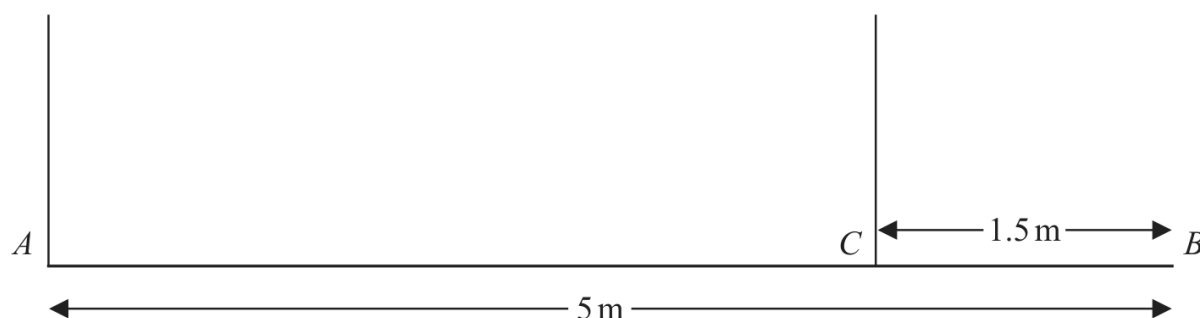
A non-uniform rod AB , 5 m, weight W , rests on supports at A and B . The reaction at B is 1.5 times the reaction at A . Find the distance of the centre of mass from A .

PAST PAPERS

Q22 PAST PAPER Pearson IAL WME01 · October 2019 · Q5

M1 · TOPIC 10 8 marks

5.

**Figure 2**

A non-uniform beam, AB , has length 5 m and mass 12 kg. The beam is suspended in a horizontal position by two vertical ropes. One rope is attached to the beam at A . The other rope is attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 2. The distance of the centre of mass of the beam from A is 1.75 m. The beam is modelled as a non-uniform rod and the ropes are modelled as light inextensible strings.

A particle of mass M kg is now placed on the beam at B and the beam remains in equilibrium in a horizontal position.

(a) Find the largest possible value of M .

(3)

The particle at B is now removed and a particle of mass 15 kg is now placed on the beam at the point D , where $AD = x$ metres. The beam remains in equilibrium in a horizontal position.

Given that the tension in the rope attached to the beam at C is now twice the tension in the rope attached to the beam at A ,

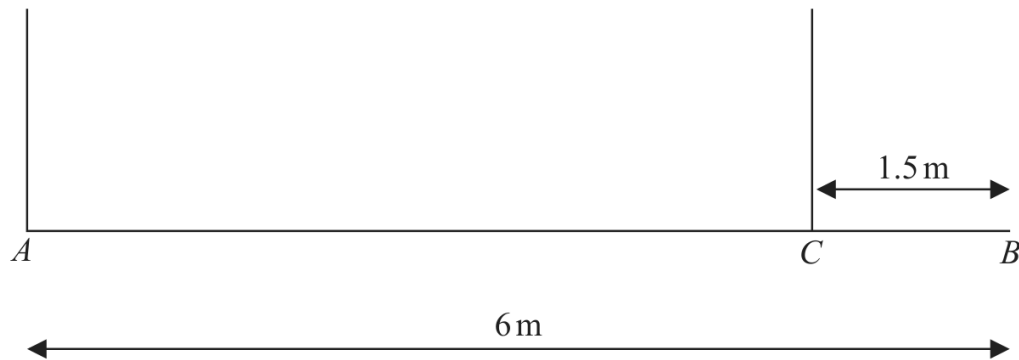
(b) find the value of x .

(5)

Q23 PAST PAPER Pearson IAL WME01 · January 2020 · Q2

M1 · TOPIC 10 7 marks

2.

**Figure 1**

A non-uniform beam AB has length 6 m and weight W newtons. The beam is supported in equilibrium in a horizontal position by two vertical ropes, one attached to the beam at A and the other attached to the beam at C , where $CB = 1.5$ m, as shown in Figure 1.

The centre of mass of the beam is 2.625 m from A .

The ropes are modelled as light strings. The beam is modelled as a non-uniform rod.

Given that the tension in the rope attached at C is 20 N greater than the tension in the rope attached at A ,

(a) find the value of W .

(6)

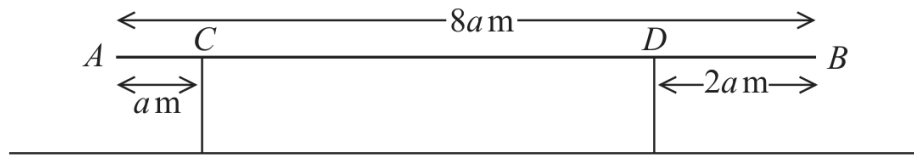
(b) State how you have used the fact that the beam is modelled as a rod.

(1)

Q24 PAST PAPER Pearson IAL WME01 · June 2021 · Q7

M1 · TOPIC 10 10 marks

7.

**Figure 2**

A non-uniform beam AB , of mass 60 kg and length $8a$ metres, rests in equilibrium in a horizontal position on two vertical supports. One support is at C , where $AC = a$ metres and the other support is at D , where $DB = 2a$ metres, as shown in Figure 2.

The magnitude of the normal reaction between the beam and the support at D is three times the magnitude of the normal reaction between the beam and the support at C .

By modelling the beam as a non-uniform rod whose centre of mass is at a distance x metres from A ,

(a) find an expression for x in terms of a .

(5)

A box of mass M kg is placed on the beam at E , where $AE = 2a$ metres.

The beam remains in equilibrium in a horizontal position.

The magnitude of the normal reaction between the beam and the support at C is now equal to the magnitude of the normal reaction between the beam and the support at D .

By modelling the box as a particle,

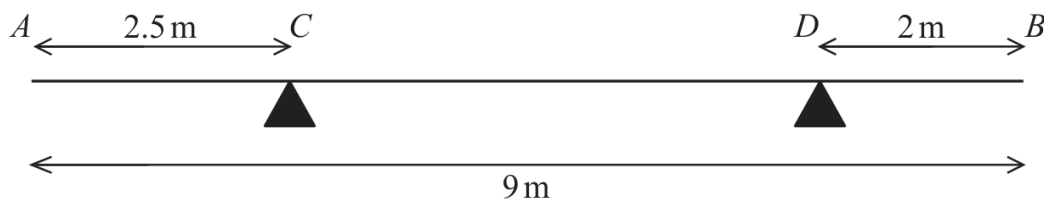
(b) find the value of M .

(5)

Q25 PAST PAPER Pearson IAL WME01 · October 2021 · Q1

M1 · TOPIC 10 6 marks

1.

**Figure 1**

A non-uniform rod AB has length 9 m and mass M kg.

The rod rests in equilibrium in a horizontal position on two supports, one at C where $AC = 2.5$ m and the other at D where $DB = 2$ m, as shown in Figure 1.

The magnitude of the force acting on the rod at D is twice the magnitude of the force acting on the rod at C .

The centre of mass of the rod is d metres from A .

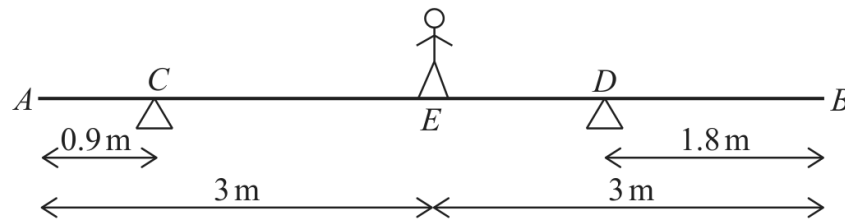
Find the value of d .

(6)

Q26 PAST PAPER Pearson IAL WME01 · October 2024 · Q2

M1 · TOPIC 10 9 marks

2.

**Figure 1**

A non-uniform beam AB has length 6 m and mass 50 kg. The beam rests horizontally on two supports at C and D , where $AC = 0.9$ m and $DB = 1.8$ m.

A child of mass 25 kg stands on the beam at E , where $AE = EB = 3$ m, as shown in Figure 1.

The beam is in equilibrium.

The magnitude of the normal reaction between the beam and the support at C is R_C newtons.

The magnitude of the normal reaction between the beam and the support at D is R_D newtons.

The beam is modelled as a rod and the child is modelled as a particle.

The centre of mass of the beam is between C and D and is a distance x metres from D .

Given that $2R_D = 3R_C$

(a) show that $x = 1.38$

(6)

The child remains at E and a block of mass M kg is placed on the beam at B .

The block is modelled as a particle.

Given that the beam is on the point of tilting,

(b) find the value of M .

(3)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|---|
| 00 | Always | Diagram with every force and every distance.
Uniform $\rightarrow$ midpoint. |
| 01 | Moment | $F \times$ perpendicular distance ($d \sin \theta$). Say the direction. |
| 02 | Equilibrium | Moments about an unknown force, then resolve. |
| 03 | Tilting | The other reaction is 0. Largest/smallest/range $\rightarrow$ tilting, reactions ≥ 0 . |
| 04 | Non-uniform | $\times$ unknown $\cdot$ two situations $\cdot$ two equations $\cdot$ divide. |

BOTTOM LINE Diagram $\rightarrow$ moments about the force you do not need $\rightarrow$ resolve $\rightarrow$ tilting means a zero reaction.

Answers

Moments • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • THE MOMENT OF A FORCE

- Q1** Dr Eslam • Perpendicular distance
 $12 \times 3 \sin 30^\circ = 18 \text{ N m}$
- Q2** Dr Eslam • Resultant with signs
 $5(2) - 3(2) = 4 \text{ N m}$ (in the sense of the 5 N force)

IDEA 02 • A ROD IN EQUILIBRIUM

- Q3** Dr Eslam • Moments about a support
 $R_C = 40 \text{ N}, R_A = 20 \text{ N}$
- Q4** Dr Eslam • Ratio of tensions
 $T_A = 2g, T_C = 8g; 8g(AC) = 10g(3) \Rightarrow AC = 3.75 \text{ m}$
- Q5** Pearson IAL WME01 • June 2023 • Q4
 (a) $p = \frac{4}{21}$ (b) $q = \frac{2}{3}$ (c) weight at the midpoint
- Q6** Pearson IAL WME01 • October 2023 • Q1
 (i) 8 N (ii) $x = \frac{7}{3}$
- Q7** Pearson IAL WME01 • January 2024 • Q5
 (b) $x = 0.08$ (c) $M = 82.5$
- Q8** Pearson IAL WME01 • January 2025 • Q4
 (b) $d = 2.5$ (c) $X = \frac{980}{3} \approx 327$
- Q9** Pearson IAL WME01 • October 2025 • Q1
 (b) $W = 24$
- Q10** Pearson IAL WME01 • January 2026 • Q4
 (a)(i) 367.5 N (ii) 122.5 N (b) $M = 195$

IDEA 03 • ON THE POINT OF TILTING

- Q11** Dr Eslam · Reaction becomes zero
- Q12** Dr Eslam · Range of a mass
- Q13** Pearson IAL WME01 · January 2021 · Q4
- Q14** Pearson IAL WME01 · January 2022 · Q3
- Q15** Pearson IAL WME01 · June 2022 · Q5
- Q16** Pearson IAL WME01 · October 2022 · Q2
- Q17** Pearson IAL WME01 · January 2023 · Q4
- Q18** Pearson IAL WME01 · June 2024 · Q4
- Q19** Pearson IAL WME01 · June 2025 · Q5
- Q20** Pearson IAL WME01 · June 2026 · Q2
- $40d = 20(1.5) \Rightarrow d = 0.75 \text{ m}$
- tilting about the support near B: $M(1) = 10(1) \Rightarrow M = 10$
 $x = 3.5$
- (a) 20 kg (b) $X = 4.05$ (c) its weight acts exactly at A
- (a) $(16.25 - M)g$ (b) $(2M - 1.25)g$ (c) $0.625 \leq M \leq 16.25$
- (a) $\frac{2}{3}M$ (b) $0.1Mg$
- (i) $W = 300$ (ii) $x = 0.63$
- (a) $M = 0.8$ (b) $5.2g = 51.0 \text{ N}$
- $d = \frac{24}{17}a$
- (a) 171.5 N (b) $X = \frac{80}{3}g \approx 261$

IDEA 04 · NON-UNIFORM RODS

- Q21** Dr Eslam · Find x from a ratio
 $R_B = 0.6W$; $0.6W(5) = Wx \Rightarrow x = 3 \text{ m}$
- Q22** Pearson IAL WME01 · October 2019 · Q5
(a) $M = 14$ (b) $x = 2.8$
- Q23** Pearson IAL WME01 · January 2020 · Q2
(a) $W = 120$ (b) the beam stays straight
- Q24** Pearson IAL WME01 · June 2021 · Q7
(a) $x = 4.75a$ (b) $M = 50$
- Q25** Pearson IAL WME01 · October 2021 · Q1
 $d = 5.5$
- Q26** Pearson IAL WME01 · October 2024 · Q2
(a) $x = 1.38$ (b) $M = 55$